

Pathogenesis of acute graft-versus-host disease: from intestinal microbiota alterations to donor T cell activation

Robert Zeiser,^{1*} Gerard Socié^{2,*} and Bruce R. Blazar^{3,*}

¹Department of Haematology, Oncology and Stem Cell Transplantation, Freiburg University Medical Centre, Freiburg, Germany,

²Haematology Stem cell transplant Unit, Saint Louis Hospital, APHP, Paris, France and ³Department of Pediatrics, Division of Blood and Marrow Transplantation, University of Minnesota, Minneapolis, MN, USA

Summary

Acute graft-versus-host disease (aGVHD) is a major life-threatening complication of allogeneic haematopoietic cell transplantation (allo-HCT). Here we discuss the aGVHD pathophysiology initiated by multiple signals that cause alloreactive T-cell activation. The outcome of such donor T-cell activation is influenced by T-cell receptor-signal strength, anatomical location, co-stimulatory/co-inhibitory signals and differentiation stage (naive, effector/memory) of T-cells. Additionally, cross-priming of T cells to antigens expressed by pathogens can contribute to aGVHD-mediated tissue injury. In addition to the properties of donor T-cell activation, highly specialized tissue resident cell types, such as innate lymphoid cells, antigen-presenting cells, immune regulatory cells and various intestinal cell populations are critically involved in aGVHD pathogenesis. The role of the thymus and secondary lymphoid tissue injury, non-haematopoietic cells, intestinal microflora, cytokines, chemokines, microRNAs, metabolites and kinases in aGVHD pathophysiology will be highlighted. Acute GVHD pathogenic mechanisms will be connected to novel therapeutic approaches under development for, and tested in, the clinic.

Keywords: graft-versus-host disease, danger associated molecular patterns, neutrophils, micro RNAs, novel targets.

Despite major advances in our understanding of the pathogenesis of graft-versus-host disease (GVHD) in recent

decades, about 50% of the patients undergoing allogeneic haematopoietic cell transplantation (alloHCT) still develop grade 2–4 acute GVHD (GVHD) and GVHD patients who are refractory to standard corticosteroid treatment have a dismal long-term prognosis with only 5–30% overall survival (Martin *et al*, 2012). Donor T-cells that destroy recipient tissues are the main mediators of GVHD, as shown by studies wherein T-cell depletion from the graft led to low levels of GVHD in humans (Horowitz *et al*, 1990), albeit associated with higher relapse rates for certain haematological malignancies and an increased risk of graft failure. Based on this central role for T-cells, a major focus of GVHD research centres upon the three signals required for T-cell activation being (i) T-cell receptor (TCR) activation, (ii) co-stimulation and (iii) cytokines, and possible therapeutic interventions at each point of signalling. T-cell activation is affected by the location and type of antigen-presenting cells (APCs), the antigen/peptide being presented and the capacity of direct recognition of foreign major histocompatibility complex (MHC)/peptides by the T-cells themselves. The classical concept, that this T-cell priming was mainly via recipient-derived dendritic cells (DCs) in secondary lymphoid organs, was recently revised when it became clear that depletion of host CD11c⁺ cells even enhanced GVHD (Koyama *et al*, 2012) and that antigen presentation to T-cells can be via recipient non-haematopoietic cells within target organs, such as myofibroblasts that express MHC class II (Koyama *et al*, 2012) and plasmacytoid DCs (Koyama *et al*, 2009) in addition to donor-derived CD103⁺ DCs that reside in the colon (Koyama *et al*, 2015). The T-cell mediated tissue damage is counterbalanced by different innate immune cells. For example, macrophages, depending on the tissue and environmental context (Hashimoto *et al*, 2011) and regulatory DCs (Sato *et al*, 2003) can protect mice from GVHD.

Besides T-cell mediated organ damage, direct cytotoxic effects caused by the release of tissue damaging factors, such as tumour necrosis factor (TNF)- α or reactive oxygen species (ROS), play a role in GVHD, particularly when GVHD evolves after conditioning regimen-related tissue damage or after an infection that triggers innate immunity. These tissue damaging effects remain connected to T-cell responses,

Correspondence: Robert Zeiser, Department of Haematology, Oncology and Stem Cell Transplantation, Freiburg University Medical Centre, Germany, Universitätsklinikum Freiburg, Hugstetter Str. 55, 79106 Freiburg, Germany.

E-mail: robert.zeiser@uniklinik-freiburg.de

Bruce R. Blazar, Department of Pediatrics, Division of Blood and Marrow Transplantation, MMC 366, 420 Delaware St SE, Minneapolis, MN 55455, USA.

E-mail: blaza001@umn.edu

*Equal contributors.

providing points of potential therapeutic interventions as will be described below.

The objectives of this review are to discuss recent insights into the pathogenesis of aGVHD and point out that, while early research focused entirely on understanding and targeting T-cells as the classical effector cells in GVHD, more recent research has extended this view to regulatory mechanisms within the target tissue itself, the influence of the microbiome, the central role of intestinal immune cell activation and metabolites produces by commensal microbiota.

T-cell receptor activation during the early phase of GVHD

The first step for a T-cell to become alloreactive is the activation of its T-cell receptor (TCR) via foreign MHC or minor antigens/peptides. In the mouse model this can be achieved by using MHC class I and/or class II strain combinations activating CD8 and/or CD4 T-cells, respectively. This 'full MHC mismatch' is not required for GVHD as patients with a human leucocyte antigen 10/10 allele matched unrelated donor also can develop GVHD, indicating that T-cell responses to minor antigen(s) can lead to the generation of alloreactive T-cells and GVHD. In the mouse model, minor antigen mismatched strain combinations exist and alloreactive T-cell responses to a single antigen mismatch can be measured via TCR transgenic T-cells recognizing either host restricted H-Y antigens when using female donors and male recipients (Toubai *et al*, 2012) or host peptides presented in the context of allo-MHC antigens (CD4: TEa; CD8: 2C). TEa transgenic CD4 T-cells express a TCR specific for a host MHC class II I-E^d derived peptide when presented within the MHC class II I-A^b molecule as would be observed when using H2^b donors and semi-allogeneic H2^{bxd} recipients (Koyama *et al*, 2012, 2015). For H2^b 2C CD8 TCR transgenic T-cells, allorecognition can occur in H2^d recipients that express the endogenous p2Ca peptide presented in the context of H2-L^d. These TCR transgenic T-cells are helpful tools to distinguish the cell populations that activate donor T-cells, in contrast to T-cells that are not directly alloreactive. Both anti-H-Y TCR transgenic T-cells and combined CD4 TEa and CD8 2C donor T-cells each resulted in >90% lethality by 2 months (Gonzalez *et al*, 2002; Toubai *et al*, 2012). On a per cell basis, recipient APCs are 100–1000 times more potent in priming allogeneic T cells compared to donor APCs (Koyama *et al*, 2012). However, by using depleting different DC subtypes and B-cells, three independent groups have shown that host haematopoietic-derived APCs are dispensable for aGVHD (Koyama *et al*, 2012; Li *et al*, 2012; Toubai *et al*, 2012). Depletion of host CD11c⁺ cells even enhanced GVHD (Koyama *et al*, 2012) and the authors showed that antigen presentation to T-cells can be via recipient non-haematopoietic cells that express MHC class II such as myofibroblasts (Koyama *et al*, 2012). Additionally, donor-derived CD103⁺ DCs that reside in the colon (Koyama *et al*,

2015) were shown to prime donor T-cells. The authors concluded that recipient APCs drive the disease, but in their absence, host non-haematopoietic APCs can act as a substitute. The T-cell mediated tissue damage is counterbalanced by different innate immune cells. For example, macrophages, depending on the tissue and environmental context (Hashimoto *et al*, 2011), and regulatory DCs (Sato *et al*, 2003) can protect mice from aGVHD. While B cells as APCs seem to be dispensable for aGVHD (Li *et al*, 2012), chronic GVHD is mediated by an interplay between B cells and T cells which can be targeted via inhibition of B-cell receptor downstream events, such as spleen tyrosine kinase (Flynn *et al*, 2015).

Intracellular signalling in T-cells is critical to support effector T cells (T_{eff}) that cause GVHD. For example, the extracellular signal-regulated kinases, Erk1/2 phosphorylation and signal and activator of transcription 3 (STAT-3) are upregulated in T-cells during GVHD and interference with STAT3 phosphorylation can inhibit GVHD (Lu *et al*, 2008). Recently we described that the strength of the TCR signal determines the fate of the alloreactive donor T cells (Schönle *et al*, 2016). We found that clustering of the TCR with the tyrosine kinase, Lck, that associates with CD4 and CD8 cytoplasmic tails, is impaired in T-cells lacking caveolin-1. Caveolin-1 (CAV1) is required for the organization of cellular microdomains and thereby influences the clustering of TCR-Lck. Consecutively reduced TCR signalling in CAV1 knock-out T-cells determines that the donor T-cells preferentially develop into regulatory T-cells (T_{reg}) and GVHD is reduced (Schönle *et al*, 2016). This was consistent with findings in other models where the strength of TCR signalling is related to the generation of T_{reg} *in vivo*, with low TCR signals due to low density or low affinity of the antigen or to premature termination of the APC:T-cell interaction, directly correlated with the conversion to T_{reg} *in vivo* in the periphery (Kretschmer *et al*, 2005; Gottschalk *et al*, 2010; Gabryšová *et al*, 2011).

T_{eff} alloresponses are also affected by thymus function. For example, it has been long known that acute GVHD incidence and severity increases in older adults compared to children. Thymic involution that occurs with aging has a major adverse effect on the thymic microenvironment especially for the thymic epithelial cells (TECs) essential for thymopoiesis (Min *et al*, 2007). Loss of TECs may lead to failed negative selection of donor anti-host reactive T-cells. TECs support the generation of donor thymic-derived T_{reg}, CD4⁺25⁺FoxP3⁺ T-cells capable of suppressing GVHD following their adoptive transfer in rodents (Cohen *et al*, 2002; Hoffmann *et al*, 2002; Taylor *et al*, 2002) and humans (Brunstein *et al*, 2011, 2016; Di Ianni *et al*, 2011; Martelli *et al*, 2014) at the time of transplant. Post-transplant, radiation and GVHD-induced thymic injury can result in failed negative selection of anti-host alloreactive T-cells and decreased thymic output of T_{reg} derived from donor haematopoietic progenitor cells, probably contributing to higher GVHD rates in older patients.

Besides these studies elucidating the basic biology of TCR activation during an allogeneic immune response, more than three decades ago it was recognized that the inhibition of a TCR downstream signal, namely calcineurin activation, would confer a high degree of T_{eff} cell immune suppression. TCR engagement leads, via several intermediate steps, to activation of calcineurin, which dephosphorylates and thereby activates the transcription factor nuclear factor of activated T-cell, cytoplasmic (NFATc). Activated NFATc is then translocated into the nucleus, where it leads to increased transcription of the *IL2* gene, which ultimately stimulates the growth and differentiation of T-cells. Calcineurin inhibitors include ciclosporin A (CyA) that binds cyclophilin, and tacrolimus (FK-506) and pimecrolimus that each bind FKBP12 proteins in T-cells. In the mouse model, reduction of *IL2* production by CyA adversely affected both T_{eff} and T_{reg} function (Zeiser *et al*, 2006, 2008a; Zeiser & Negrin, 2008b). In contrast, another FKBP12 target drug, the mammalian target of rapamycin (also termed the mechanistic target of rapamycin, mTOR) permitted relatively preserved T_{reg} function since T_{reg} are less dependent than T_{eff} cells on the mTOR/Akt pathway (Zeiser *et al*, 2006, 2008a; Zeiser & Negrin, 2008b). CyA became an important component of GVHD prophylaxis in the 1980s (Storb *et al*, 1986), whereas tacrolimus was further developed in the 1990s for GVHD and graft rejection prophylaxis (Blazar *et al*, 1994a). Combined tacrolimus and methotrexate (MTX) after unrelated alloHCT in patients (Ratanatharathorn *et al*, 1998) was shown to significantly decrease the risk for acute GVHD but not overall and relapse-free survival rates compared to CyA and MTX (Ratanatharathorn *et al*, 1998).

Besides the standard immunosuppressive drugs, cyclophosphamide (CP) given on day 2 after transplantation in murine bone marrow chimera induced tolerance towards skin allografts (Mayumi & Good, 1989). Based on these proof-of-concept studies, post-transplant CP was applied to mice developing acute GVHD and was shown to reduce the disease by depleting alloreactive T cells while relatively sparing T_{regs} (Ganguly *et al*, 2014). Clinical trials revealed that post-transplant CP is effectively maintaining human T_{reg} in aGVHD prophylaxis (Kanakry *et al*, 2014). While CyA and tacrolimus are the backbone of most immunosuppressive regimens for patients currently undergoing allo-HCT worldwide, there is increasing use of CP-based and mTOR inhibitor-based regimens. Besides CP for T cell depletion, TCRalpha/beta-CD19 lymphocyte depletion (that spares TCRgamma/delta cells) for the prophylaxis of acute GvHD has been successfully used (Lang *et al*, 2015).

The balance between costimulation and co-inhibition of T-cell activation

The second crucial event after the TCR has been activated is costimulation, which alone is typically insufficient to trigger productive T-cell responses. The absence of costimulatory

signals leads to T-cell anergy (antigen-specific hypo- or non-responsiveness upon antigenic re-stimulation) or apoptosis. CD28, a member of the immunoglobulin (Ig) superfamily, is the receptor for CD80 (B7.1) and CD86 (B7.2) proteins and the only B7 receptor constitutively expressed on naive T-cells. As described below, cytotoxic T-lymphocyte associated protein-4 (CTLA4; CD152), induced with T-cell activation, is a high avidity receptor that competes with CD28 for CD80 and CD86 binding, resulting in downregulation of T-cell responses. The concept of using CD28 blocking reagents in the setting of GVHD was tested in the 1990s when it was shown that CTLA4-Ig (Blazar *et al*, 1994b; Wallace *et al*, 1994) and anti-B7 blocking monoclonal antibodies (mAbs) (Blazar *et al*, 1996) reduced GVHD lethality in rodents. Tolerance induction to host alloantigens also could be achieved *in vitro* in murine or human allo- mixed-lymphocyte reaction cultures with CTLA4-Ig (Blazar *et al*, 1996). In a first-in-human *in vivo* trial in alloHCT patients, blockade of CD28 mediated costimulation with abatacept (protein comprised of the high avidity extracellular domain of CTLA4 fused to an IgG1 tail to prolong half-life) was associated with low GVHD rates (Koura *et al*, 2013). In contrast, a superagonist of CD28 (TGN1412) that was studied in patients with autoimmunity as immunosuppressive drug caused a severe cytokine storm and multiple organ failure. Explanations have included the fact that superagonistic anti-CD28 mAb also may stimulate other cells, such as eosinophilic granulocytes leading to the release of *IL2*, *IL4*, *IL13* and γ -interferon (IFN- γ) (Woerly *et al*, 1999), a function similar to bacterial superantigen responses bypassing the need for TCR signalling (Horvath *et al*, 2012) or work in conjunction with weak or tonic TCR signals (Hunig, 2012).

Beside blockade of B7 ligand family members, several members of the Ig/CD28 supergene family have proven to be potent inhibitors of GVHD in preclinical models. For example, other T-cell expressed costimulatory molecules are upregulated during GVHD such as the inducible T-cell costimulator (ICOS; CD278), found on activated T-cells, can be targeted to prevent GVHD in rodent models (Hubbard *et al*, 2005; Taylor *et al*, 2005). Similarly, several members of the TNF/TNF-receptor family, including CD40 ligand (CD40LG; CD154) (Blazar *et al*, 1997a), 4-1BB (CD137; TNFRSF9) (Blazar *et al*, 2001), CD30 (TNFRSF8) (Blazar *et al*, 2004) and OX40 (CD134; TNFRSF4) (Blazar *et al*, 2003a) are upregulated on alloreactive T-cells and blockade of these receptor/ligand interactions can be highly effective in preventing GVHD lethality in preclinical rodent models.

The natural signalling opponents of the costimulatory molecules are the co-inhibitory molecules. The first well-studied co-inhibitory molecule was CTLA4 (Brunet *et al*, 1987; Walunas *et al*, 1994). Other co-inhibitory molecules and their ligands that are often induced by proinflammatory mediators, released during conditioning regimen injury and by alloreactive T-cells, have been identified and studied in GVHD. For example, programmed death protein-1 (PD1;

PDCD1) binds to PD-L1 (CD274) and PD-L2 (PDCD1LG2) that are upregulated during GVHD, reducing the severity of GVHD (Blazar *et al*, 2003b). Similarly, T-cell immunoglobulin and mucin-domain containing protein 3 (TIM3; HAVCR2) and its ligand, galectin-9 (LGALS9), work in concert to lessen GVHD lethality (Veenstra *et al*, 2012). Other molecules upregulated on T-cell upon activation including B- and T-lymphocyte attenuator (BTLA)(Sakoda *et al*, 2011), lymphotoxin-like compete with herpes simplex virus glycoprotein D for HVEM, a receptor expressed by T lymphocytes (LIGHT;TNFSF14) (Xu *et al*, 2007), lymphocyte-activation gene 3 (LAG3) (Sega *et al*, 2014), T-cell immunoreceptor with Ig and ITIM domains (TIGIT) (Yu *et al*, 2009; Levin *et al*, 2011), B7-H3 (Veenstra *et al*, 2015) and B7-H5 (PD-1H; VISTA, V-domain Ig suppressor of T cell activation; C10orf54) (Flies *et al*, 2015) reduce GVHD lethality in rodent models. Since several co-stimulatory and co-inhibitory molecules are expressed on T_{eff} and T_{reg} as well as APCs, careful interpretation should be given to studies based upon the loss of expression of either donor or host cells as this restricted loss of function may not be translatable to studies using mAbs or proteins that would affect both donor and host cells.

While in GVHD the dampening of the immune response is warranted under other conditions, such as relapse, immune activation is the major therapeutic goal. In preclinical studies, it was also shown that CD28/B7 interactions are required for sustaining the graft-versus-leukaemia (GVL)

effects (Blazar *et al*, 1997b), highlighting the need for monitoring GVL responses in the clinic. In other studies, an antagonistic anti-CTLA4 mAb (Blazar *et al*, 1999) and anti-PD-1/ligand blocking agents (Asakura *et al*, 2010; Koestner *et al*, 2011; Michonneau *et al*, 2016) increased the GVL effects. In this clinical setting, application of the antagonistic anti-CTLA4 antibody ipilimumab led to anti-tumour responses in some patients with refractory malignancies that had failed disease control by allo-HCT (Bashey *et al*, 2008; Davids *et al*, 2016). Similarly, blocking PD-1/PD-1 ligands in patients with Hodgkin lymphoma, including 78% that had failed autologous HCT, produced response rates of ~80% (Ansell *et al*, 2015). Based upon the impressive preclinical and clinical data, allo-HCT studies using PD-1 blocking reagents for anti-tumour immune responses are now in progress. The multiple molecules involved in co-stimulation and co-inhibition of T-cell activation are summarized in Fig 1. Several of these different co-stimulatory and co-inhibitory pathways have been investigated in murine bone marrow transplant (BMT) models such as: co-stimulatory: CD28, CD80/CD86, OX40, CD40LG, ICOS, and co-inhibitory: CTLA4, VISTA, PD-1, PD-L1, TIGIT and LAG3 pathways.

Cytokines that promote T-cell activation and survival

The third signal required for sustained T-cell activation is provided by cytokines, initially ascribed to IL12. Cytokines

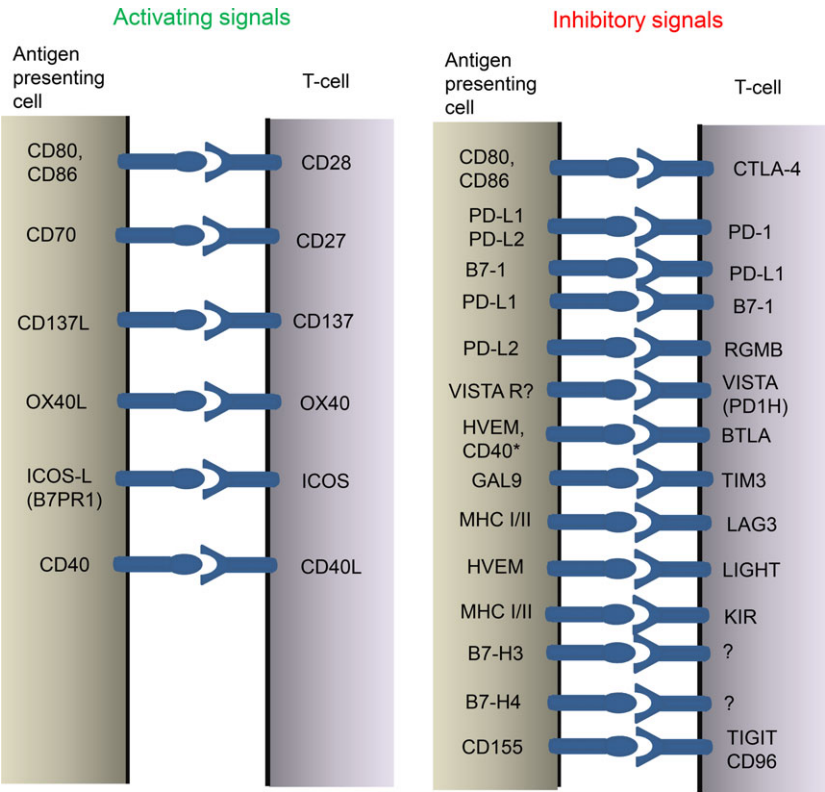


Fig 1. Displayed are exemplary costimulatory and coinhibitory ligand/receptor pairs on the antigen presenting cells and T cells. ICOS, inducible T-cell co-stimulator; ICOS-L, ICOS-ligand; CTLA-4, Cytotoxic T-lymphocyte-associated antigen 4; PD-1, Programmed death protein-1; BTLA, B- and T-lymphocyte attenuator; LAG3, Lymphocyte-activation gene 3; TIGIT, T cell immunoreceptor with Ig and ITIM domains; TIM3, T cell immunoglobulin and mucin-domain containing protein 3; RGMB, repulsion guidance molecule b; VISTA/ PD1H, V-domain Ig suppressor of T cell activation; HVEM, herpes virus entry mediator A. *BTLA has multiple binding partners. Of these different co-stimulatory and co-inhibitory pathways the following have been investigated in murine and human transplant studies: co-stimulatory: CD28, CD80/CD86, OX40, CD40L, ICOS, and co-inhibitory: CTLA4, VISTA, PD-1, PD-L1, TIGIT and LAG3.

also are crucial of T-cell survival, especially the IL2 receptor common gamma chain (CD132; IL2RG) cytokines IL2, IL7 and IL15. Based on the importance of these cytokines we recently applied inhibition of the common gamma chain by an antibody and found that this intervention potently reduced GVHD in the mouse model (Hechinger *et al*, 2015). Blocking CD132 reduced downstream signals in T-cells including Janus activated kinase-3 (JAK3) and, consistent with a role of this molecule, we found that JAK3-deficient T-cells were subjected to significantly less GVHD (Hechinger *et al*, 2015). Besides common gamma chain cytokines, multiple other cytokines have JAK1/2 activation as a common nominator. Consistently, inhibition of JAK1/2 by ruxolitinib was shown by different groups to reduce GVHD in the mouse model (Spoerl *et al*, 2014; Carniti *et al*, 2015). A recent study suggests that topical ruxolitinib also suppresses GVHD and protects skin follicular stem cells in mice, whereas topical corticosteroids, commonly used for treatment of skin in aGVHD patients, inhibit skin stem cells and niche pre-adipocytes (Takahashi *et al*, 2016). To translate these findings in the clinic, several centres used ruxolitinib as an individual approach to treat patients with steroid-refractory GVHD (Zeiser *et al*, 2015). The overall response rate (ORR) was 82% (44/54) in steroid-refractory aGVHD, including 25 complete responses (46%), while for steroid-refractory chronic GVHD the ORR was 85% (35/41) (Zeiser *et al*, 2015). Ruxolitinib will be investigated in a prospective trial in Germany (NCT02396628) and a clinical trial using the JAK1 selective inhibitor INCB39110 has begun for the treatment of GVHD (NCT02614612). Major differences between selective JAK1 compared to JAK1 and JAK2 inhibition include a potentially lower risk of cytopenia when only JAK1 is inhibited, though potentially at a reduced efficacy for GVHD reduction as JAK2 inhibition or JAK2 deficiency in donor T-cells alone reduced GVHD (Betts *et al*, 2015). Other cytokine-based approaches that can have efficacy in preclinical rodent models include neutralizing or preventing the production of proinflammatory cytokines, such as IL6, as shown in mice (Chen *et al*, 2009; Tawara *et al*, 2011) and in preliminary studies in allo-HCT patients (Kennedy *et al*, 2014), the IL12p40 subunit that is shared with IL12 and IL23 (Das *et al*, 2010), IL21 (Bucher *et al*, 2009; Hanash *et al*, 2011; Hippen *et al*, 2012) and IL33 (Reichenbach *et al*, 2015; Zhang *et al*, 2015). Notably, IL33 has both potent pro- and anti-inflammatory properties as IL33 administration administered before and shortly after the conditioning expands recipient T_{reg} that results in protecting mice against aGVHD (Matta *et al*, 2016). Overall, the use of anti-cytokine therapies or T-cell costimulatory blockers for aGVHD prophylaxis has to be considered with caution, as these novel agents could be associated with a loss of GVL effect. In contrast to the T cell activation that occurs after conditioning-induced tissue damage, donor lymphocyte infusions (DLIs) can induce GVHD at late time points. DLI-induced GVHD differs from conventional GVHD as it evolves often in the absence of tissue damage

due to conditioning, presumably leading to a different degree of T cell activation by pro-inflammatory cytokines.

Chemotaxis of T-cells towards GVHD target organs

After naïve T-cells have received the optimal three signals, which are delivered most often in secondary lymphoid organs, those T-cells that are alloreactive need to migrate to GVHD target tissues to mediate injury and receive further signals for their differentiation and to sustain their longevity. To subvert this step of GVHD pathophysiology induction, chemokines that guide the migration of T-cells towards GVHD target organs have been targeted in mouse GVHD models [reviewed in (Wysocki *et al*, 2005)]. While some investigators found chemokine receptor blockade to be effective (Murai *et al*, 1999, 2003), this strategy may be context-dependent as radiation conditioning regimen therapy can overcome directed tissue migration and trapping of donor T-cells that occurs as a result of an individual chemokine/receptor interaction. One example is CCR5 inhibition that was protective against GVHD in a non-irradiated GVHD mouse model (Murai *et al*, 1999) but GVHD onset was earlier and severity worsened when CCR5^{-/-} T cells were transferred into recipients receiving total body irradiation (TBI) (Welniak *et al*, 2004; Wysocki *et al*, 2004; Koura *et al*, 2013). Conversely, a single institution phase-I trial reported that CCR5 inhibition prevents GVHD of liver and gut before day 100 (Reshef *et al*, 2012), while recent data using CCR5 inhibition in the setting of reduced-intensity conditioning revealed no protection from GVHD (Hammond *et al*, 2016). The Blood and Marrow Transplant Clinical Trials Network Trial 1203 (NCT02208037) that prospectively investigated whether CCR5 inhibition reduces GVHD has recently finished accrual. Data from this trial as well as future studies involving multi-centre trials under different conditioning regimen intensities are needed to better understand whether CCR5 blockade will prove advantageous in reducing GVHD incidence or severity.

Besides CCR5, the sphingosine 1-phosphate receptor (S1PR) has been recognized as a therapeutic structure to affect immune cell migration. T-cell egress from the lymph node (Kim *et al*, 2003) and DC migration (Taylor *et al*, 2007) were potently inhibited by the S1PR agonist FTY720 in murine GVHD (Taylor *et al*, 2012). Based on these findings the S1PR agonist KRP-203 is currently evaluated in a clinical study on patients undergoing allo-HCT. In addition to S1PR, danger-associated molecular patterns, such as uric acid or adenosine triphosphate, shown to be involved in GVHD (Wilhelm *et al*, 2010; Jankovic *et al*, 2013; Klämbt *et al*, 2015), may contribute to the migration of DCs, inflammatory monocytes or T-cells towards inflamed or damaged tissues. The multiple factors that affect activation, migration and cytolytic function of donor T-cells during GVHD are summarized in Fig 2. The net effect of positive *versus*

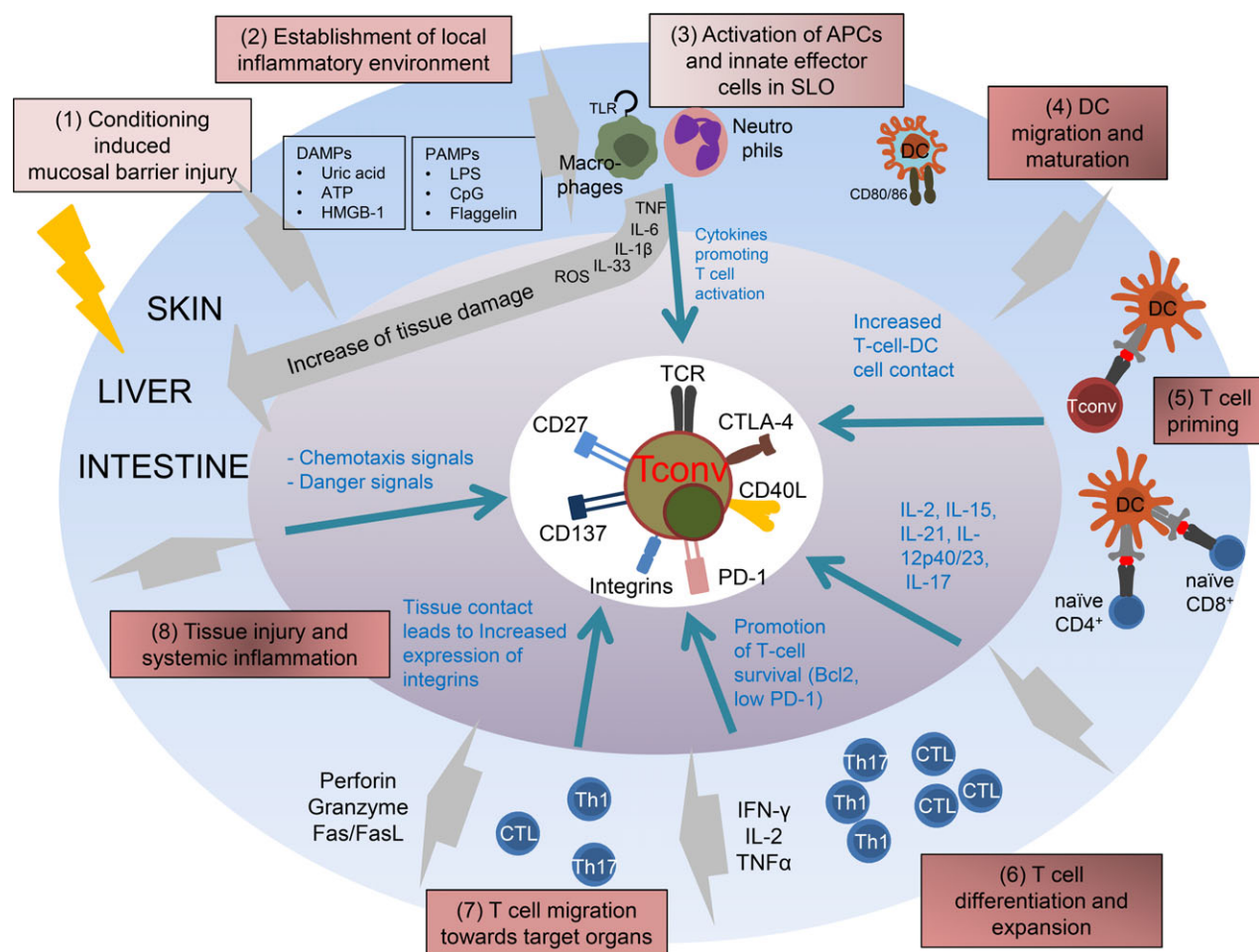


Fig 2. The different soluble factors and environmental cues that promote T-cell activation during the different steps from initial tissue injury to the clinical picture of acute graft-versus-host disease are shown. The steps include amongst others: (1) Conditioning-induced mucosal barrier injury, (2) Establishment of a local inflammatory environment, (3) Activation of APCs and innate effector cells in secondary lymphatic organs, (4) DC migration and maturation, (5) T-cell priming, (6) T-cell differentiation and expansion, (7) T cell migration towards target organs, (8) Tissue injury and systemic inflammation. APC, antigen-presenting cell; ATP, Adenosine triphosphate; CTL, cytotoxic T-lymphocyte; DAMPs, danger associated molecular patterns; DC, dendritic cell; HMGB-1, High mobility group box 1 protein; IL, interleukin; LPS, lipopolysaccharide; PAMPs, pathogen associated molecular patterns; ROS, reactive oxygen species; SLO, secondary lymphatic organs; Tconv, conventional T cell; TCR, T-cell receptor; Th, T-helper cell; TNF, tumour necrosis factor.

negative signals that act on T cells (Fig 2 and 3) determines their cytokine pattern, survival, migratory activity and cytotoxic activity leading to the clinical picture of GVHD or protection from GVHD.

Microbiome, metabolites and immune regulation in aGVHD biology

The intestinal tract and its microbiome were recognized as central determinants of GVHD initiation in seminal work performed in the 1970s, when it was reported that germ-free mice have virtually no GVHD risk (van Bakkum & Knaan, 1977). However, clinical therapeutic interventions to use this knowledge were not overall successful as recreating a germ-free intestinal environment in patients with an established

microbiome was fraught with substantial challenges. Broad-spectrum antibiotics were even shown to increase GVHD-related mortality in patients and mice (Shono *et al*, 2016). Laboratory mice are housed in most institutions in specific pathogen-free (SPF), though not germ-free, barrier facilities. A recent study shows that standard laboratory mouse husbandry has profound effects on the immune system (Beura *et al*, 2016). Co-housing of laboratory mice with free-living barn populations of feral mice and pet store mice produced mice with immune systems closer to those of adult humans with a predilection toward T_{eff} cells and memory T-cell populations distributed in the intestinal mucosa (Beura *et al*, 2016). Such co-housing resulted in global changes in blood cell gene expression of the laboratory mice towards patterns that more closely reflected the immune signatures of adult

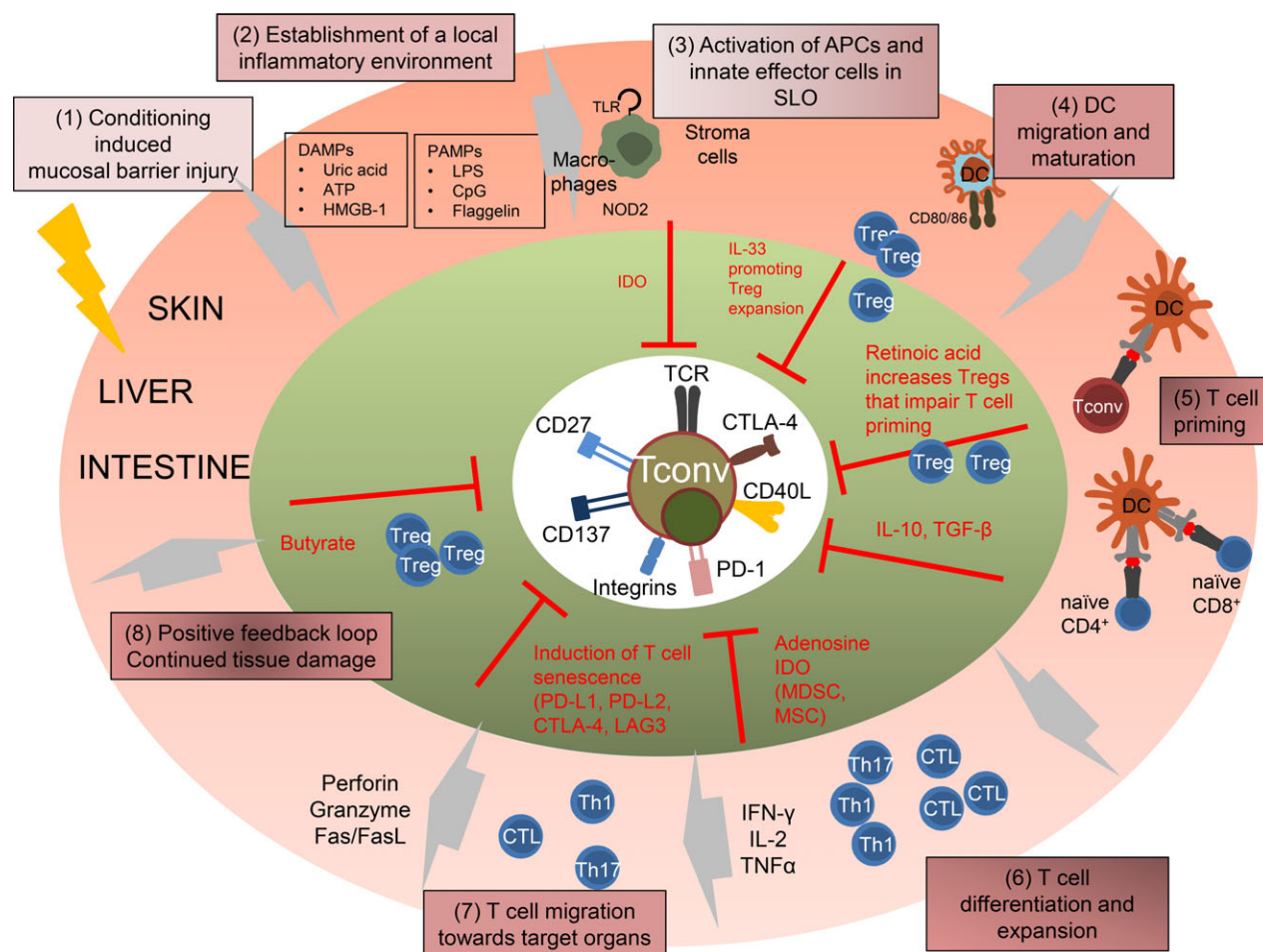


Fig 3. Inhibitory signals that counteract T cell activation in the context of acute GVHD are shown under the same conditions as depicted in Fig 2. APC, antigen-presenting cell; ATP, Adenosine triphosphate; CTL, cytotoxic T-lymphocyte; DAMPs, danger associated molecular patterns; DC, dendritic cell; HMGB-1, High mobility group box 1 protein; IDO, indoleamine 2,3-dioxygenase; IFN, interferon; IL, interleukin; LPS, lipopolysaccharide; MDSC, myeloid derived suppressor cells; MSC, mesenchymal stroma cells; NOD2, nucleotide binding oligomerization domain containing 2; PAMPs, pathogen associated molecular patterns; SCFA, short chain fatty acid; SLO, secondary lymphatic organs; Tconv, conventional T cell; TGF, transforming growth factor; Th, T-helper cell; TLR, Toll-like receptor; TNF, tumour necrosis factor; Treg, regulatory T cell.

humans, thus suggesting the 'dirtier' mice have more intrinsic activation that should lead to more GVHD (Beura *et al*, 2016).

Several studies indicate that neutrophils and inflammatory monocytes are recruited to the intestinal tract where they cause tissue damage and thereby promote disease development (Socié *et al*, 2004; Giroux *et al*, 2011; Schwab *et al*, 2014; Klämbt *et al*, 2015). The recruitment of neutrophils was abrogated when germ-free mice were used as allo-HCT recipients. Neutrophils lacking multiple Toll-like receptors caused less severe tissue damage while anti-apoptotic BclX-transgenic neutrophils that had a prolonged survival enhanced GVHD (Schwab *et al*, 2014). The tissue injury in intestinal GVHD is characterized by severe villous atrophy and crypt degeneration (Wall *et al*, 1971; Sale *et al*, 1979; Epstein *et al*, 1980). The principal cell types of the epithelium of the small intestine are Paneth cells, intestinal stem cells (ISCs), goblet cells, enterocytes and enteroendocrine

cells. The regenerative potential of the intestinal tract relies on ISCs, which are in close contact and interaction with Paneth cells that also reside within the bottom of the crypts. Besides secreting factors that help sustain and modulate the epithelial stem and progenitor cells in the intestines, Paneth cells secrete antimicrobial peptides and inflammatory mediators that protect from pathogens and shape the commensal microbiota. When exposed to bacteria or their products (e.g. lipopolysaccharide, muramyl dipeptide and lipid A), Paneth cells start to secrete defensins, which insert into bacterial membranes, where they interact with one another and induce the formation of pores that disrupt membrane function, ultimately leading to cell lysis (Ayabe *et al*, 2000). Paneth cells also release lysozymes (Rocha *et al*, 2015), TNF-α (Keshav *et al*, 1990), and phospholipase A2 (Senegas-Balas *et al*, 1984) with antimicrobial activity. It has been shown that ISCs and Paneth cells are targets in GVHD in mice (Takashima *et al*, 2011; Eriguchi *et al*, 2012; Hanash *et al*, 2012)

while irradiation targets primarily ISCs (Takashima *et al*, 2011). Factors that have been shown to have the potential to enhance the regeneration of ISC are interleukin-22 (IL22), R-spondin-1 (RSPO1) and keratinocyte growth factor (KGF; fibroblast growth factor-7, FGF7) amongst others. ISCs express IL22 receptors and recent studies highlight the fundamental role of IL22 proteins produced by innate lymphoid cells type-3 (ILC3s) for the protection of the ISC (Hanash *et al*, 2012; Lindemans *et al*, 2016). The location of IL22 secreting cells in close proximity to the ISC is, most likely, key for its potent protective effects as others have shown that exogenous IL22 given systemically to mice enhanced GVHD (Zhao *et al*, 2014). IL22 deficiency of the recipient increases crypt apoptosis, loss of ISCs, and GVHD mortality (Hanash *et al*, 2012). While IL22 administration may be a promising strategy to protect ISCs (Dudakov *et al*, 2015) and is to be tested in the treatment of lower gastrointestinal GVHD (NCT02406651), the ideal timing and dosing is so far unclear as exogenous IL22 also can be pro-inflammatory under certain conditions.

Another potentially gut-protective or regenerative protein is KGF, which can be produced by intraepithelial TCR gamma/delta cells and mesenchymal cells. KGF is not selective for ISC but rather for different intestinal epithelial cells. Exogenous KGF administration, designed to increase epithelial cell numbers prior to radiation, reduced GVHD in mouse models (Panoskaltis-Mortari *et al*, 1998), although the survival benefit of mice varied between the different reports ranging from modest (Panoskaltis-Mortari *et al*, 1998) to very potent anti-GVHD effects (Krijanovski *et al*, 1999). Based on rodent GVHD as well as a phase III randomized autologous HCT trial (Spielberger *et al*, 2004) studies, KGF (Palifermin, Kevivance™) was applied to patients in a phase I/II randomized placebo controlled trial where it did not reduce acute GVHD but did lower mucositis rates in patients conditioned with more intensive TBI *versus* chemotherapy regimens (Blazar *et al*, 2006). The latter mucositis benefits were noted in the aforementioned phase III autologous HCT trial (Spielberger *et al*, 2004; Stiff *et al*, 2006).

In other studies, Wnt signalling has been shown to promote the regenerative potential of ISC and the Wnt ligand RSPO1 promoted intestinal integrity and survival of mice suffering from GVHD. Administration of RSPO1 to mice undergoing allogeneic BMT reduced GVHD by protecting ISCs from conditioning injury (Takashima *et al*, 2011). Mechanistically, RSPO1 stimulated ISCs to differentiate into secretory cells and inhibited GVHD-associated microbiome changes most likely by promoting Paneth cell secretion of anti-microbial peptides as described above. The protective effect of RSPO1 was connected to the microbial microflora in the intestinal tract and partly abrogated when mice were treated with broad-spectrum antibiotics. Another highly specialized intestinal cell type is the goblet cell, which produces mucin that shields the epithelium from luminal bacteria (Pelaseyed *et al*, 2014). Consistently, goblet cell injury in

GVHD reduces the mucous layer of the colon and permits translocation and dissemination of dominant luminal pathogens in mice developing GVHD (Penack *et al*, 2010). The observation that Paneth cell loss and intestinal microbial dysbiosis are hallmarks of the early phase of GVHD has been found in mice and patients (Jenq *et al*, 2012; Holler *et al*, 2014; Taur *et al*, 2014).

Besides exogenous manipulation of intestinal cells, a major focus of research has been devoted to the intestinal microbiome and its products. Recent elegant work showed that intestinal microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate GVHD (Mathewson *et al*, 2016). Intestinal microbiota metabolize the nutrients ingested by the host and release products of their own metabolism. Therefore the intestinal metabolome is a composition of products from the mammalian metabolism of host enterocytes, ISC, Paneth cells and goblet cells and microbial metabolites from the luminal bacteria. Intestinal homeostasis depends heavily on microbiota-derived metabolites. The microbial metabolites affect both non-immune cells and immune cells. For example selected strains of *Clostridia* that are known to produce the short chain fatty acid (SCFA) butyrate result in increased frequencies of T_{reg} in the intestinal tract (Atarashi *et al*, 2013). This links the intestinal metabolism to GVHD as T_{reg} can potentially suppress GVHD (Cohen *et al*, 2002; Hoffmann *et al*, 2002; Taylor *et al*, 2002). Butyrate, secreted by intestinal bacteria, is an histone deacetylase (HDAC) inhibitor, shown to potentially reduce GVHD by modulating indoleamine-2,3-dioxygenase (IDO)-dependent innate immune and allo-stimulating functions of APCs in a STAT3-dependent manner (Reddy *et al*, 2008; Choi & Reddy, 2014). IDO itself is upregulated on intestinal parenchymal cells and APCs in rodents (Jasperson *et al*, 2008) and in patients with GVHD (Ratajczak *et al*, 2012), probably as a result of IFN- γ produced by alloreactive T-cells (Jasperson *et al*, 2009). High IDO expression results in the depletion of the essential amino acid tryptophan, resulting in metabolic starvation and apoptosis of donor T-cells via a stress response mechanism. Moreover, high IDO expression favours immune suppression (Munn *et al*, 2002) along with T_{reg} generation and function in mice (Sharma *et al*, 2007) and humans (Chen *et al*, 2008) (Fig 3). Aside from direct immune regulatory activities of IDO, such as T-cell apoptosis or peripheral T_{reg} generation, tryptophan catabolites may directly suppress the GVHD response via their immune suppressive properties (Jasperson *et al*, 2009). Consistently, the microbiome-derived metabolite butyrate was shown to be protective against GVHD and its deficiency enhanced intestinal epithelial cell damage due to GVHD (Mathewson *et al*, 2016) (Fig 3). Composition of the ingested food also plays a role in regulating rapid changes in the composition of the gut microbiota (De Vadder *et al*, 2014) as, for example, the amount of carbohydrates may affect the mucous layer and yoghurt may promote the amount of GVHD-protective lactobacilli (Gerbitz *et al*, 2004; Jenq *et al*, 2012).

Thus, these data from different investigators indicate that the microbial metabolome impacts intestinal immune homeostasis and thereby also GVHD.

These studies draw attention to the metabolic requirements of highly proliferative T_{eff} cells and the energy needs for immune suppressive T_{reg} in offsetting GVHD pathogenesis. Highly proliferating T-cells and T_{reg} utilize multiple energy sources to keep pace with metabolic demands during GVHD. There is general agreement that aerobic glycolysis is essential for optimal GVHD T_{eff} responses (Gatza *et al*, 2011; Saha *et al*, 2013; Nguyen *et al*, 2016); in some studies, T_{eff} cells also have been shown to utilize oxidative phosphorylation and fatty acid oxidation (Gatza *et al*, 2011; Byersdorfer *et al*, 2013; Saha *et al*, 2013) and glutaminolysis (Nguyen *et al*, 2016), supplying vital energy needs not met by glycolysis alone. Thus, targeting metabolic pathways and subverting T-cell energy utilization via essential amino acid deprivation or inhibitors of glycolysis, fatty acid oxidation and oxidative phosphorylation, and/or glutaminolysis may reduce the frequency of rapidly proliferating T_{eff} cells that are required to mediate GVHD tissue injury.

Another approach against GVHD that affects metabolic pathways is inhibition of 3-hydroxy-3-methyl-glutaryl-coenzyme A reductase (HMG-CoA reductase; HMGCR), the rate-limiting enzyme of the L-mevalonate pathway. A group of drugs called statins inhibit HMG-CoA reductase and thereby deplete farnesyl- and geranylgeranyl-residues in T cells and APCs (Hechinger *et al*, 2013). In mouse models of aGVHD, statins were shown to reduce GVHD severity without a loss of GVH effects (Zeiser *et al*, 2007). Based on these data from the mouse model, subsequent clinical trials using statin prophylaxis in both patients/donors (Hamadani *et al*, 2013) or using treatment of donors only (Rotta *et al*, 2010) reported low GVHD rates. Conversely, one negative trial was reported where donors and patients received statins as prophylaxis leading to low rates of grade III-IV aGVHD but unusually high rates of isolated upper intestinal tract acute GVHD (Efebera *et al*, 2016). A pharmacological approach targeting the proteasome with bortezomib was shown to reduce aGVHD in the mouse model by reducing pro-inflammatory cytokines (Sun *et al*, 2004). Clinical studies using a bortezomib-based GVHD prophylaxis yielded low aGVHD rates (Koreth *et al*, 2012).

Additional immune regulatory mechanisms in the intestinal tract include the release of IL33 from intestinal stromal cells and T-cells. IL33 signalling of its receptor, ST2 (IL1R1) on donor T-cells in the context of proinflammatory cytokines, such as IL12 or IL18, can drive T-cell alloresponses and tissue destruction (Yang *et al*, 2011; Reichenbach *et al*, 2015). Conversely, under non-inflammatory conditions, IL33 augments the proliferation and increases the number of gut T_{reg} and myeloid-derived suppressor cells that can inhibit GVHD lethality (Highfill *et al*, 2010; Matta *et al*, 2016). Similarly, retinoic acid has opposing functions in driving the stability (Brown *et al*, 2015) and generation of Th1 T_{eff} cells that cause GVHD (Aoyama *et al*, 2013) as well as a co-factor

in the generation of murine (Benson *et al*, 2007) and human (Golovina *et al*, 2011) T_{reg} . Thus, the net results of gut IL33 and retinoic acid release on immune responses and regulation in the intestine of GVHD mice is dictated by the timing and nature of the inflammatory environment in the intestine.

microRNAs in GVHD

High throughput analysis, such as those that use proteomics, microarrays, whole exosome sequencing and MicroRNAs (miRs) arrays, applied to human samples are increasingly used as discovery approaches that may provide new candidate molecules with a role in GVHD pathophysiology. MiRs are small endogenous noncoding RNAs that post-transcriptionally regulate gene expression by base pairing with sequences typically located in the 3' untranslated region (3' UTR) of target messenger RNAs (mRNAs) and subsequently repress translation or induce mRNA degradation (Liu, 2008). MiRs are potent regulators of multiple target genes and their systematic analysis has led to the identification of miR clusters that are either pro- or anti-inflammatory. Studies on blood samples of patients undergoing allo-HCT showed that certain miRs were strongly connected to the risk of GVHD development and may serve as valuable non-invasive biomarkers for aGVHD (Xiao *et al*, 2013; Gimondi *et al*, 2016), as demonstrated for miR-155 (MIR155) and miR-146a (MIR146A) (Xie *et al*, 2014; Atarod *et al*, 2016). Different miRs were shown to be functionally relevant for GVHD including miR-155 (Ranganathan *et al*, 2012; Chen *et al*, 2015), miR-100 (MIR100) (Leonhardt *et al*, 2013), miR-146a (Stickel *et al*, 2014), miR-142 (MIR142) (Sun *et al*, 2015) and miR-17-92 (MIR17HG) (Wu *et al*, 2015).

A first miR that was functionally linked to GVHD was miR-155 (Ranganathan *et al*, 2012). The authors showed that mice receiving miR-155-deficient T cells experienced reduced GVHD-related mortality compared to those receiving wild type T cells. Consistently, GVHD was more aggressive in recipients of miR-155 overexpressing T-cells (Ranganathan *et al*, 2012). These findings were connected to a potential therapeutic approach, as blocking miR-155 via a synthetic anti-miR-155 decreased GVHD severity in mice receiving wild type T-cells. Analyses of human specimens revealed miR-155 up-regulation in the presence of intestinal GVHD (Ranganathan *et al*, 2012). Compatible with a role in T-cell immune responses, miR-155 had been shown to positively regulate the function and lineage commitment of pathogenic T-cells in murine experimental autoimmune encephalomyelitis (Murugaiyan, 2011). Given that miR-155 regulates not only T-cells but also the activation of the innate immune system, a second study analysed its function in DCs during GVHD (Chen *et al*, 2015). Deficiency of miR-155 in the recipient was connected to improved survival and reduced serum levels of pro-inflammatory cytokines (Chen *et al*, 2015). By using miR-155-deficient bone marrow chimeric mice receiving allo-HCT and miR-155-deficient DCs, the

authors then showed that miR-155 deficiency in the host DC compartment was responsible for protection from GVHD. Mechanistically, activated miR-155-deficient DCs displayed lower expression of various purinergic receptors and impaired migration toward adenosine triphosphate (Chen *et al*, 2015) and exhibited multiple defects, including mitogen-activated protein kinase pathway dysregulation and reduced inflammasome-associated gene expression (Chen *et al*, 2015). Functionally, caspase-1 cleavage, and IL1 β (IL1B) production in miR-155-deficient DCs were reduced upon activation. Consistent with the *in vitro* data, NLRP3 inflammasome/miR-155 double-knockout allo-HCT recipient mice had no increased protection from GVHD compared with NLRP3-deficient recipients (Chen *et al*, 2015), indicating a redundancy of function of miR-155 deficiency in the context of miR-155 deficiency.

Another miR that controlled T-cell activation during GVHD was miR-142 (Sun *et al*, 2015). By using mice with a targeted deletion of mir-142, the authors showed that T-cell proliferation was reduced after stimulation both *in vitro* and in multiple murine GVHD models (Sun *et al*, 2015). Mechanistically, miR-142-deficient T-cells demonstrated substantial cell-cycling defects, due to upregulation of genes involved in cell cycling. *E2f7* and *E2f8*, two predicted miR-142 target genes upregulated during cell cycling, were most highly upregulated in miR-142-deficient cells and their silencing in miR-142-deficient T-cells ameliorated cell-cycling defects and reduced GVHD (Sun *et al*, 2015).

In another study, transfer of miR-146a-deficient T-cells caused increased GVHD severity, elevated TNF- α serum levels, and reduced survival (Stickel *et al*, 2014). The authors found that TNF receptor-associated factor 6 (TRAF6), a verified target of miR-146a, was increased in miR-146a-deficient T-cells upon contact with alloantigen. The augmented TRAF6 levels translated into increased nuclear factor- κ B activity and TNF- α production in miR-146a-deficient T cells (Stickel *et al*, 2014). MiR-146a overexpression using a miR-146a mimic reduced GVHD severity. The minor genotype of the single nucleotide polymorphism (SNP) rs2910164 reduces expression of miR-146a. Consistent with the findings in mice, the presence of this SNP in HCT donors was associated with severe aGVHD (grade III/IV) (Stickel *et al*, 2014). However, clinical trials employing simple TNF- α antagonists as prophylaxis were negative (Hamadani *et al*, 2008). *MIR17HG* (miR-17-92 cluster), which encodes 6 miRs including 17, 18a, 19a, 20a, 19b-1, and 92-1 was shown to be critically involved in the ability of donor T-cells to induce GVHD (Wu *et al*, 2015) by promoting CD4 T-cell activation, proliferation, survival, and Th1 differentiation, while inhibiting Th2 and inducible T_{reg} differentiation (Wu *et al*, 2015). As miRs act on the expression of multiple target genes and may thereby influence the immune system at different cellular and functional levels, they are potentially attractive targets for the modification of allogeneic immune responses using miR mimics and inhibitors.

Summary and outlook

While GVHD remains a significant clinical problem, the understanding of its biology has improved significantly over the past 3 decades. It is clear that T-cells are the major effector cells causing GVHD and multiple strategies aimed at inhibiting the three signals (TCR activation, co-stimulation and proinflammatory cytokines) that T cell activation requires have been extensively investigated. While inhibition of co-stimulation was shown to be potent at reducing GVHD, recent success of checkpoint inhibitors point towards a use of these antibodies for enhancing GVL effects. Besides the TCR signal and co-stimulation, cytokines and chemokines are required for full T-cell activation and to allow for their migration towards target tissues. Based on findings in mice, strategies that block cytokines, such as IL6, or chemokines, such as CCR5, were already transferred into the clinical application in early trials. While these strategies target only individual molecules, the inhibition of certain kinases may have a more potent effect as the signal of multiple cytokine or chemokine receptors may be prevented as, for example, with Rho-associated kinase 2 (Flynn *et al*, 2016) inhibition or JAK1/2 inhibition (Zeiser *et al*, 2015). However, although T-cells have been a focus of translational research aiming at reducing GVHD, tissue-protective strategies with KGF, RSPO1 or IL22 are under intensive investigation. More recently, more global approaches, such as targeting modification of acetylation, SCFA metabolism, the microbiome or miRs, have become promising strategies to prevent GVHD. Overall, these individual strategies hold promise to reduce GVHD, possibly in a combinatorial approach in order to make allo-HCT more safe for patients with haematological malignancies.

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Author contribution

R.Z., G.S. and B.R.B. collected literature, discussed the studies and contributed equally to the writing of the manuscript.

Conflict of interest

The authors have no conflict of interest to disclose.

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